GREENDALE · STR SOLUTIONS LTD · INTERNAL GUIDE

Fresh PC theke *backend* chalu kora

tested on BullMQ 6.5.3

Windows PC te PostgreSQL 18, Redis 8 (WSL2), Prisma 7 ar BullMQ 6 setup, prothom migration, ar shob kichu kaj korche kina test.

Version	Kar jonno	Shomoy
v1 · 31 Oct 2026	Sifat (hands-on DB)	~60–90 min

Kon command kothay chalaben, label dekhe bujhben: POWERSHELL Windows terminal UBUNTU WSL terminal SQL psql ba pgAdmin Query Tool PROJECT VS Code terminal, `greendale-back` folder e

00 Ki ki install hobe

TOOL	VERSION	KENO
Node.js	24 LTS	Prisma 7 er minimum 20.19. 24 LTS shob library te tested.
PostgreSQL	18 (latest 18.6)	Ekhon er latest stable. 19 ekhono RC, production e nibo na.
pgAdmin 4	installer er shathe	Compass er jaygay. Free.
WSL2 + Ubuntu	24.04 LTS	Redis er official build Linux e. VPS o Ubuntu, tai dev ar prod same.
Redis	8.x (official apt repo)	BullMQ er backend. Ubuntu er default apt e purono 7.0 ache, oita nibo na.
Redis Insight	latest	Redis er GUI, queue key dekhar jonno. Free, official.
Prisma	`7.10.1` (exact pin)	ORM + migration.
BullMQ / ioredis	`6.5.3` / `6.0.0`	Job queue. Ei combination container e chalie test kora.

MongoDB theke PostgreSQL: ki kothay gelo

MONGODB E	POSTGRESQL E
Compass	pgAdmin 4 / DBeaver / `npx prisma studio`
mongodb+srv://... URI	`postgresql://user:pass@localhost:5432/greendale?schema=public`
Atlas M0	Lagbe na. Local e install, prod e VPS e nijer Postgres. Shob free, khoroch shudhu VPS er.
Collection / document	Table / row. Shape fixed thake `schema.prisma` te
Schema change kora lage na	Protita change e ekta migration file hoy, git e commit hoy
mongodump / mongorestore	`pg_dump` / `pg_restore`

> **EKTA TRAP AGE THEKE JENE RAKHUN**
>
> npm e `prisma` package er `latest` tag ekhon **8.0.0-rc** (release candidate). Shudhu `npm i prisma` dile RC install hobe, kintu `@prisma/client` 7.10.1 e thakbe, ar version mismatch hobe. Tai sob shomoy exact version diye install korben (Step 6 e command ache).

01 Prerequisite: virtualization, Node, editor

1A. BIOS e virtualization on kora

WSL2 ar Docker duitar jonnoi lage. Apnar PC te eta ekhon off ache (remote shell chalu korte giye ei error eshechilo).

1. Age check korun: **Task Manager → Performance → CPU**. Nicher dike **Virtualization: Enabled** dekhale 1A skip korun.
2. PC restart dite din, ar boot er shomoy BIOS key chepe dhore rakhun. Key brand onujayi alada hoy (beshir bhag e `Del`, `F2` ba `F10`). Windows theke shortcut: Settings → System → Recovery → Advanced startup → UEFI Firmware Settings.
3. Option ta khujun: Intel e **Intel Virtualization Technology (VT-x)**, AMD te **SVM Mode**. Shadharonoto Advanced ba CPU Configuration menu te thake. **Enabled** kore Save & Exit korun.
4. Windows e fire abar Task Manager check korun, ekhon Enabled dekhabe.

1B. Node.js 24 LTS

1. `nodejs.org` theke **Windows Installer (.msi), 24.x LTS** namie install korun. "Tools for native modules" checkbox tick dite hobe na.
2. Notun PowerShell khule check korun:

```powershell
node -v      # v24.x.x
npm -v
```

1C. VS Code extension

- **Prisma** (official): `schema.prisma` e syntax highlight, format ar error dekhay.
- **WSL** (Microsoft): Ubuntu er file VS Code theke edit korar jonno. Optional.

02 PostgreSQL 18 install

1. `postgresql.org/download/windows` page theke **"Download the installer"** e click korun (EDB). **18.x, Windows x86-64** nin.
2. Installer chalan. Installation directory default thakuk: `C:\Program Files\PostgreSQL\18`.
3. **Components:** PostgreSQL Server ✓, pgAdmin 4 ✓, Command Line Tools ✓. **Stack Builder** er tick tule din, eta lagbe na.
4. **Data directory** default thakuk.
5. **Password:** eta `postgres` superuser er password. Strong password din, password manager e rakhun. Eta app e use korbo na, shudhu admin kaj er jonno.
6. **Port:** `5432` (default). Ager kono Postgres theke thakle installer 5433 suggest korbe, tokhon `.env` eo 5433 likhben.
7. **Locale:** "[Default locale]" thakuk. Windows e UTF-8 encoding je kono locale er shathe kaj kore.
8. Next → Install. Shesh e "Launch Stack Builder" er tick tule Finish korun.

2A. `psql` ke PATH e add kora

1. Start menu te **"Edit the system environment variables"** khujun → Environment Variables.
2. System variables e **Path** → Edit → New → `C:\Program Files\PostgreSQL\18\bin` → OK.
3. PowerShell bondho kore abar khulun, tarpor check korun:

```powershell
psql --version                # psql (PostgreSQL) 18.x
Get-Service postgresql*       # Status: Running, StartType: Automatic
```

Service ta Windows boot er shathe nijei chalu hoy. Bondho thakle `services.msc` e giye **postgresql-x64-18** → Start.

03 App er jonno user ar database

Mongo te shorashori ekta URI diye kaj korten. Postgres e niyom holo, app **superuser diye connect korbe na**. App er jonno alada ekta role (user) thakbe, ar database tar nijer (owner).

```powershell
psql -U postgres -h localhost      # Step 2 er superuser password chabe
```

```sql
CREATE ROLE greendale WITH LOGIN PASSWORD 'gd_local_2026';
ALTER ROLE greendale CREATEDB;     -- local only, see note below
CREATE DATABASE greendale
  OWNER greendale
  ENCODING 'UTF8'
  TEMPLATE template0;

\l greendale                       -- check: Owner = greendale, Encoding = UTF8
\q
```

PROTTEKTA LINE KENO

CREATEDB: `prisma migrate dev` protibar ekta temporary "shadow database" banay, change gulo safe kina dekhar jonno. Ei permission na thakle error ashbe `P3014`. VPS e ei permission dibo na, karon okhane shudhu `migrate deploy` chole.

OWNER greendale: PostgreSQL 15 theke `public` schema te shudhu database er owner table banate pare. Owner na hole migration e error ashe *permission denied for schema public*.

UTF8 + template0: Bangla nam (student, guardian) thik moto save hobe, ar sorting o thik hobe.

Connection string ta ei rokom hobe (Step 6 e `.env` e boshaben):

```
DATABASE_URL="postgresql://greendale:gd_local_2026@localhost:5432/greendale?schema=public"
```

PASSWORD E SPECIAL CHARACTER

Password e `@ : / # ? %` thakle URL e encode kore likhte hobe (`@` → `%40`, `#` → `%23`). Nahole Prisma `P1000` / `P1013` error dey. Local e shudhu letter ar digit diye password rakhai shohoj.

04 Data dekha (Compass er motoi)

pgAdmin 4 (installed)

1. pgAdmin khulun. Prothom bar ekta master password chaite pare, din.
2. Left panel e **Servers** e right click → Register → Server.
3. **General:** Name e likhun `Greendale local`.
4. **Connection:** Host `localhost`, Port `5432`, Database `greendale`, Username `greendale`, password din, "Save password" tick din.
5. Table dekhte: Databases → greendale → Schemas → public → Tables → right click → **View/Edit Data**.

Prisma Studio (shobcheye druto)

Project folder theke:

```
PROJECT
npm run db:studio
```

Browser e table gulo khulbe, Compass er moto edit o kora jay. Daily kaj e eta-i beshi kaje lagbe.

DBeaver (optional)

Free, ekta app diye Postgres, MySQL, SQLite shob chole. VPS er DB dekhar shomoy **SSH tunnel** tab ta kaje lagbe (Section 10 dekhun).

05 Redis 8: WSL2 Ubuntu te

Redis er official Windows build nai. Industry standard holo Windows dev machine e WSL2 er bhetore Redis chalano. Windows theke `localhost:6379` diye shorashori connect hoy.

5A. WSL2 + Ubuntu

```
POWERSHELL (RUN AS ADMINISTRATOR)
wsl --install -d Ubuntu-24.04
```

1. Restart chaile restart din.
2. Ubuntu nije theke khulbe. Ekta Linux username (choto hater, space chara) ar password set korun. Ei password `sudo` er jonno lagbe.
3. Check korun: `wsl -l -v` chalale **Ubuntu-24.04 · VERSION 2** dekhabe.

5B. Official Redis repo theke install

```
UBUNTU
sudo apt-get update
sudo apt-get install -y lsb-release curl gpg

curl -fsSL https://packages.redis.io/gpg | sudo gpg --dearmor -o /usr/share/keyrings/redis-archive-keyring.gpg
sudo chmod 644 /usr/share/keyrings/redis-archive-keyring.gpg
echo "deb [signed-by=/usr/share/keyrings/redis-archive-keyring.gpg] https://packages.redis.io/deb $(lsb_release -cs) main" \
  | sudo tee /etc/apt/sources.list.d/redis.list

sudo apt-get update
sudo apt-get install -y redis
redis-server --version              # v=8.x
```

5C. BullMQ er jonno config (must)

```
UBUNTU
sudo nano /etc/redis/redis.conf
```

Nicher line gulo khuje bodle din. Line na thakle file er shesh e add korun. Nano te khujte `Ctrl+W`, save korte `Ctrl+O`, ber hote `Ctrl+X`.

```
bind 127.0.0.1 -::1
requirepass gd_redis_local
maxmemory 256mb
maxmemory-policy noeviction
appendonly yes
appendfsync everysec
```

UBUNTU

```
sudo systemctl enable redis-server
sudo systemctl restart redis-server

redis-cli -a gd_redis_local ping                              # PONG
redis-cli -a gd_redis_local config get maxmemory-policy      # noeviction
```

KENO EI SETTING GULO

noeviction: Redis er memory bhore gele default policy key mucche jayga khali kore. BullMQ er job o ashole Redis key, tai SMS ba result job kono error chara harie jabe. `noeviction` dile Redis notun write e error dey, ar amra sheta log e dekhte pai. BullMQ docs e eta "very important" bola ache.

appendonly: restart ba crash hole queue te thaka job harabe na.

requirepass: prod e eta lagbei, tai local eo same rakhchi jate code e kono "dev only" path na thake.

WSL IDLE HOLE BONDHO HOYE JAY

Kono Ubuntu terminal khola na thakle Windows kichukkhon por WSL bondho kore dite pare, tokhon Redis o bondho. Tai dev shuru korar age Start menu theke **Ubuntu** ekbar khule rakhun. systemd Redis nijei chalu kore debe. Worker e `ECONNREFUSED 127.0.0.1:6379` dekhale age eta check korben.

5D. Redis Insight (GUI)

`redis.io/insight` theke Windows installer namie install korun. Add database → Host `127.0.0.1`, Port `6379`, Password `gd_redis_local`. Queue chalu hole `gevs:email:*` er moto key dekhben.

Virtualization kono bhabei on kora na gele fallback hishebe **Memurai Developer** ache (Windows native, Redis-compatible). Dev edition e kichu limitation ache, tai eta shudhu temporary hishebe use korben.

06 Project wiring: Prisma 7

6A. Install (exact version)

PROJECT

```
cd E:\Works\Greendale-School\greendale-back
npm init -y
npm i -E @prisma/client@7.10.1 @prisma/adapter-pg@7.10.1 pg bullmq@6.5.3 ioredis@6.0.0 dotenv
npm i -D -E prisma@7.10.1
```

JS-ONLY RULE ER JONNO EKTA DECISION

Prisma 7 er notun `prisma-client` generator **.ts file** generate kore. Kono setting diye JS output pawa jay na. Amader rule holo "no TypeScript anywhere", tai amra `prisma-client-js` generator use korchi. Prisma 7 e eta deprecated, kintu fully supported, ar client ager motoi `node_modules` e generate hoy. Prisma 8 stable hole abar dekhte hobe.

Prisma 7 e ar duita jinish bodleche. **Driver adapter** (`@prisma/adapter-pg`) ekhon mandatory. Ar Prisma nije `.env` load kore na, tai `prisma.config.js` e `dotenv` lage.

6B. package.json (relevant part)

```
package.json
{
  "type": "module",
  "engines": { "node": ">=22.12" },
  "scripts": {
    "dev": "node --watch --env-file=.env src/server.js",
    "dev:worker": "node --watch --env-file=.env src/worker.js",
    "db:migrate": "prisma migrate dev && prisma generate",
    "db:deploy": "prisma migrate deploy",
    "db:generate": "prisma generate",
    "db:studio": "prisma studio",
    "db:seed": "prisma db seed",
    "check:db": "node --env-file=.env scripts/check-db.js",
    "check:queue": "node --env-file=.env scripts/queue-smoke.js"
  }
}
```

App process e `.env` load kore Node er nijer `--env-file` flag. `dotenv` shudhu Prisma CLI er jonno.

6C. .env

```
.env  (git e jabe na; .gitignore e add korun)
NODE_ENV=development
PORT=4000
TZ=Asia/Dhaka
DATABASE_URL="postgresql://greendale:gd_local_2026@localhost:5432/greendale?schema=public"
REDIS_URL="redis://:gd_redis_local@127.0.0.1:6379"
QUEUE_PREFIX=gevs
```

6D. prisma.config.js (project root)

```
prisma.config.js
import 'dotenv/config';
import { defineConfig, env } from 'prisma/config';

export default defineConfig({
  schema: 'prisma/schema.prisma',
  migrations: {
    path: 'prisma/migrations',
    seed: 'node prisma/seed.js',
  },
  datasource: {
    url: env('DATABASE_URL'),
  },
});
```

6E. Schema: test korar jonno ekta table

prisma/schema.prisma

```
generator client {
  provider = "prisma-client-js"
}

datasource db {
  provider = "postgresql"   // url ekhon prisma.config.js e, ekhane na
}

model HealthCheck {
  id        Int      @id @default(autoincrement())
  note      String
  createdAt DateTime @default(now()) @db.Timestamptz(6)
}
```

6F. Prisma client (adapter shoho)

src/core/prisma.js

```
import { PrismaClient } from '@prisma/client';
import { PrismaPg } from '@prisma/adapter-pg';

const adapter = new PrismaPg({
  connectionString: process.env.DATABASE_URL,
  max: 10, // pool size; one VPS, two processes (api + worker)
});

export const prisma = new PrismaClient({ adapter });
```

6G. Prothom migration + test

PROJECT

```
npm run db:migrate -- --name init
# creates prisma/migrations/2026..._init/migration.sql, applies it, generates client
```

scripts/check-db.js

```
import { prisma } from '../src/core/prisma.js';

const [row] = await prisma.$queryRaw`
  SELECT version() AS version,
         current_database() AS db,
         current_user AS "user",
         current_setting('server_encoding') AS encoding,
         current_setting('TimeZone') AS timezone`;
console.table(row);

const created = await prisma.healthCheck.create({ data: { note: 'hello from Greendale' } });
console.log('Inserted row:', created);

await prisma.$disconnect();
```

PROJECT

```
npm run check:db
# expect: PostgreSQL 18.x · greendale · greendale · UTF8, then an inserted row
```

Ekhon `npm run db:studio` khulle **HealthCheck** table e row ta dekhben. Ei porjonto hole PostgreSQL setup shesh ✓

TIMEZONE NIYE

Local Postgres hoyto Asia/Dhaka dekhabe, kintu VPS UTC te thakbe. Tai code kokhono DB server er timezone er upor depend korbe na. Timestamp gulo `Timestamptz` e, ar attendance er `DATE` app e Asia/Dhaka diye hishab kore pathano hobe (PRD rule).

07 BullMQ: queue ar worker

Process duita alada: API (`server.js`) job **add** kore, ar worker (`worker.js`) job **process** kore. Duita alada PM2 process hishebe chole, tai ekta PDF render hote 10 second laglei API slow hoy na.

src/core/queue.js

```
import { Queue } from 'bullmq';
import IORedis from 'ioredis';

export const QUEUE_PREFIX = process.env.QUEUE_PREFIX || 'gevs';

export const QUEUES = Object.freeze({
  SMS: 'sms',
  EMAIL: 'email',
  PDF: 'pdf',
  RESULT: 'result',
});

// Producer side (API process): fail fast when Redis is down,
// so a request never hangs waiting for Redis.
const producerConnection = new IORedis(process.env.REDIS_URL, {
  enableOfflineQueue: false,
});
producerConnection.on('error', (err) => console.error('[redis:producer]', err.message));

const defaultJobOptions = {
  attempts: 3,
  backoff: { type: 'exponential', delay: 30_000 },
  removeOnComplete: { age: 7 * 24 * 3600, count: 5000 },
  removeOnFail: { age: 30 * 24 * 3600 },
};

const queues = new Map();

/** @param {string} name one of QUEUES */
export function getQueue(name) {
  if (!queues.has(name)) {
    const queue = new Queue(name, {
      connection: producerConnection,
      prefix: QUEUE_PREFIX,
      defaultJobOptions,
    });
    queue.on('error', (err) => console.error(`[queue:${name}]`, err.message));
    queues.set(name, queue);
  }
  return queues.get(name);
}

export async function closeQueues() {
  await Promise.allSettled([...queues.values()].map((q) => q.close()));
  await producerConnection.quit();
}
```

src/worker.js

```
import { Worker } from 'bullmq';
import IORedis from 'ioredis';
import { QUEUES, QUEUE_PREFIX } from './core/queue.js';

// Worker side: maxRetriesPerRequest MUST be null (BullMQ requirement).
const connection = new IORedis(process.env.REDIS_URL, { maxRetriesPerRequest: null });

const processors = {
  [QUEUES.SMS]: async (job) => {
    console.log('[sms] would send', job.data);
  },
  [QUEUES.EMAIL]: async (job) => {
    console.log('[email] would send', job.data);
  },
};

const workers = Object.entries(processors).map(([name, processor]) => {
  const worker = new Worker(name, processor, {
    connection,
    prefix: QUEUE_PREFIX,
    concurrency: 5,
  });
  worker.on('completed', (job) => console.log(`[${name}] done #${job.id}`));
  worker.on('failed', (job, err) => console.error(`[${name}] failed #${job?.id}: ${err.message}`));
  worker.on('error', (err) => console.error(`[${name}] worker error:`, err.message));
  return worker;
});

console.log(`Worker up: ${Object.keys(processors).join(', ')} (prefix ${QUEUE_PREFIX})`);

let shuttingDown = false;
async function shutdown(signal) {
  if (shuttingDown) return;
  shuttingDown = true;
  console.log(`${signal}: finishing active jobs...`);
  await Promise.allSettled(workers.map((w) => w.close()));
  await connection.quit();
  process.exit(0);
}
process.on('SIGINT', () => shutdown('SIGINT'));
process.on('SIGTERM', () => shutdown('SIGTERM'));
```

7A. Smoke test

```
scripts/queue-smoke.js
import { getQueue, closeQueues, QUEUES } from '../src/core/queue.js';

const queue = getQueue(QUEUES.EMAIL);
await queue.waitUntilReady();

const job = await queue.add('smoke-test', {
  to: 'you@example.com',
  subject: 'BullMQ smoke test',
  at: new Date().toISOString(),
});
console.log(`Queued job #${job.id}. Check the worker terminal.`);

const counts = await queue.getJobCounts('waiting', 'active', 'completed', 'failed');
console.table(counts);

await closeQueues();
```

Duita terminal lagbe:

```
TERMINAL 1
npm run dev:worker          # Worker up: sms, email (prefix gevs)

TERMINAL 2
npm run check:queue         # Queued job #1 ...

# Terminal 1 e dekhabe:
[email] would send { to: 'you@example.com', ... }
[email] done #1
```

Ei output ta container e asol PostgreSQL ar Redis er bipore chalie mila hoyeche (BullMQ 6.5.3 + ioredis 6.0.0). Worker e Ctrl+C dile age `finishing active jobs...` dekhabe, tarpor bondho hobe.

7B. BullMQ er rules (production e lagbe)

RULE	KENO
Worker e `maxRetriesPerRequest: null`	Na dile BullMQ start e exception throw kore.
Producer e `enableOfflineQueue: false`	Redis down hole `queue.add()` shathe shathe error dey. Request jhule thake na, ar service error ta handle korte pare.
`removeOnComplete / removeOnFail`	Default e BullMQ shob job chirodin rakhe, ar Redis memory dhire dhire bharte thake. Amra 7 din / 30 din rakhchi.
`QUEUE_PREFIX` per school	Product hishebe 10–12 school ekta Redis share korleo queue mishe jabe na (`gevs:`, `cbis:`...).
Idempotent `jobId`	Absent SMS: `jobId: \`absent-${enrollmentId}-${date}\`` . Same jobId er job already thakle BullMQ duplicate nibe na. Final guarantee DB te (unique row), queue shudhu ekta extra layer. jobId te `:` na diye `-` use korun.
Job data choto rakhun	Puro student object na, shudhu ID pathaben. Worker DB theke fresh data porbe.
PDF queue `concurrency: 1–2`	Protita Puppeteer instance ~150–300 MB RAM ney, ar VPS ekta-i.

NOTUN JINISH, JENE RAKHUN

BullMQ 6 e ekhon PostgreSQL backend o ache (Redis chara). Ta-o amra Redis e thakchi. PRD te Redis fixed, ar BullMQ er shobcheye mature, battle-tested path ekhono Redis. Pore Redis bad dite chaile code e shudhu `connection` ta bodlate hobe.

08 Daily workflow

Protidin kaj shuru

1. Start menu theke **Ubuntu** khulun (Redis chalu hobe).
2. VS Code terminal 1: `npm run dev`
3. Terminal 2: `npm run dev:worker`
4. Data dekhte: `npm run db:studio`

PostgreSQL Windows service, tai age theke-i chalu thake.

Schema change korle

1. `schema.prisma` edit korun.
2. `npm run db:migrate -- --name add_fee_ledger`
3. Generate hoya `migration.sql` ekbar pore dekhun.
4. Schema ar migration folder ekshathe commit korun.

KOKHONO KORBEN NA

Kono migration ekbar commit ba deploy hoye gele oi file edit ba delete korben na, notun migration banaben. Server e `migrate reset` ba `db push` na (reset puro data mucche fele). Local e reset kora jay, kintu tar age backup nin.

Backup ar restore (local e practice)

```powershell
mkdir backups
pg_dump -U greendale -h localhost -Fc -f backups\greendale-2026-10-31.dump greendale

# restore test: alada DB te, asol ta te na
createdb -U greendale -h localhost greendale_restore_test
pg_restore -U greendale -h localhost --no-owner -d greendale_restore_test backups\greendale-2026-10-31.dump
```

Protibar password type korte na chaile `%APPDATA%\postgresql\pgpass.conf` file e ei line ta likhun: `localhost:5432:*:greendale:gd_local_2026`

09 Error hole

ERROR	KARON → SOMADHAN
`P1000` Authentication failed	Password bhul, ba special character encode kora hoyni → `.env` er URL check korun.
`P1001` Can't reach database	Service bondho → `Get-Service postgresql*`, ba port bhul (5432 naki 5433).
`P3014` shadow database	Role e CREATEDB nai → `ALTER ROLE greendale CREATEDB;`
permission denied for schema public	Database er owner greendale na → `ALTER DATABASE greendale OWNER TO greendale;`
`@prisma/client did not initialize yet`	`npm run db:generate` chalan. Prisma 7 e migrate nije generate kore na.
Prisma CLI ar client er version mile na	RC install hoye geche → `npm i -D -E prisma@7.10.1`
`ECONNREFUSED 127.0.0.1:6379`	WSL / Redis bondho → Ubuntu khulun, `sudo systemctl status redis-server`
`NOAUTH` / `WRONGPASS`	`REDIS_URL` e password nai → format holo `redis://:PASSWORD@127.0.0.1:6379` (`:` er age username faka).
Eviction policy is ... It should be "noeviction"	5C er config apply hoyni → `redis.conf` check kore Redis restart din.
maxRetriesPerRequest must be null	Worker er connection e `{ maxRetriesPerRequest: null }` deya hoyni.
Stream isn't writeable ... enableOfflineQueue	API Redis e connect hobar agei job add korche → Redis chalu ache kina dekhun. Code e `queue.add` try/catch e rakhun.
Job "waiting" e atke ache	Worker chalu nai, ba API ar worker er `QUEUE_PREFIX` alada.

10 VPS e ki alada (preview)

Deployment Gemini PRD follow kore korbe. Tobe local er sathe parthokko gulo ekhane:

JINISH	LOCAL (WINDOWS)	VPS (UBUNTU 24.04)
PostgreSQL	EDB installer	PGDG apt repo theke `postgresql-18`
Redis	WSL2	Same `packages.redis.io` repo, same config + strong password
Port 5432 / 6379	localhost	Shudhu localhost. `ufw` e khola thakbe shudhu 22, 80, 443.
DB GUI	Shorashori	pgAdmin/DBeaver er **SSH Tunnel** diye (VPS user + key, DB host `localhost`)
Role permission	CREATEDB ache	CREATEDB nai, strong random password
Migration	`db:migrate`	Shudhu `npm run db:deploy`
Process	2 terminal	PM2: `greendale-api` + `greendale-worker`
Backup	Manual	Daily cron `pg_dump -Fc`, 14 din rakha, offsite copy, mashe ekbar restore test

Shesh checklist

- ✓ Task Manager e Virtualization: Enabled
- ✓ `node -v` 24.x, `psql --version` 18.x
- ✓ `npm run check:db` → UTF8 + inserted row
- ✓ `redis-cli … config get maxmemory-policy` → noeviction
- ✓ `npm run check:queue` → worker e "done #1"
- ✓ pgAdmin, Prisma Studio, Redis Insight teen tai connect hoy
- ✓ Ekta `pg_dump` + restore test kora hoyeche

Sources (checked 31 Oct 2026): postgresql.org (18.6 latest, 19 RC1) · Prisma 7 upgrade guide · Prisma generators · Prisma config reference · BullMQ: going to production · BullMQ: connections · Redis on Windows